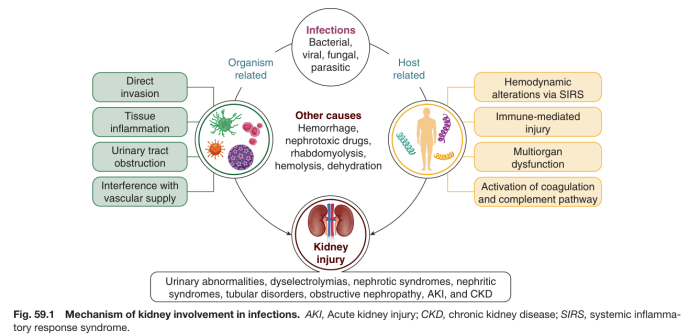
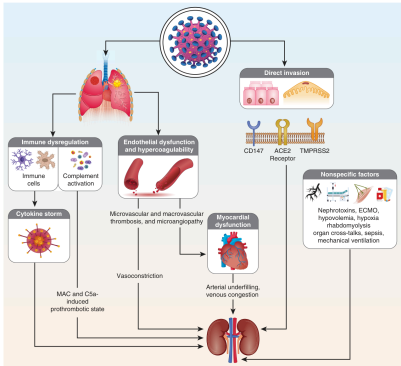




Fig_59_1



Fig_59_2

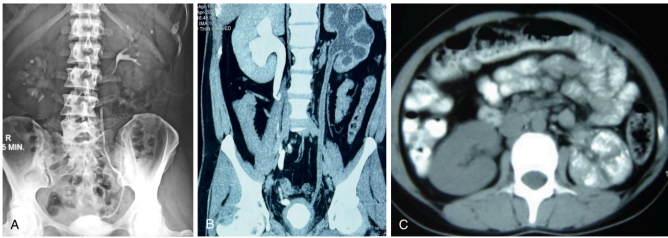


Fig. 59.3 TB radiology: radiologic abnormalities in genitourinary tuberculosis. (A) An intravenous urogram showing infundibular stenosis, pelvic scarring, irregular dilations of calyces, and cavitations of the right kidney. (B) Computed tomography (CT) urography showing destruction of parenchyma of left kidney (autonephrectomy), with thickened pelvicalyceal system and ureter. Right kidney is obstructed due to poorly compliant, fibrosed, and thickened bladder wall with reduced capacity. (C) Plain CT scan showing lobar calcifications in the left kidney. (Image courtesy Dr. Anupam Lal and Dr. Shrawan K. Singh.)

Fig_59_3

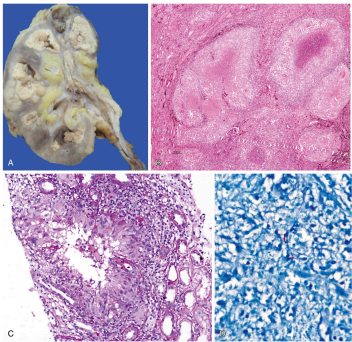


Fig. 59.4 (A) Gross image of end-stage kidney disease secondary to multiple cysts in the calyces, filled with grey-white friable putrescent material. The ureteric wall is thickened, forming megacyst. (B) Scanner and low magnification view of kidney tissue showing numerous granuloma with central necrosis, admixed with Langhans multinucleated giant cells (PAS stain). (C) Core biopsy of the kidney shows epithelioid cell granuloma with central breakdown and Langhans giant cells (PAS, 20x). (D) Acid-fast bacillus in renal tissue (Ziehl-Neelsen stain, 100x). (Image courtesy Dr. Mahesha Vankalakunti.)

Fig_59_4

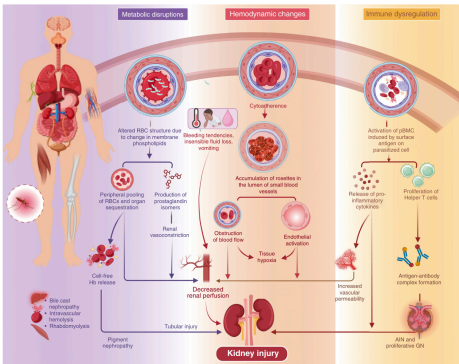


Fig. 59.5 Pathogenesis of kidney injury in malaria. AN, Acute interstitial nephritis; GN, glomerulonephritis; AB, hemoglobin; PMBC, peripheral blood mononuclear cells; RBCs, red blood cells.

Fig_59_5

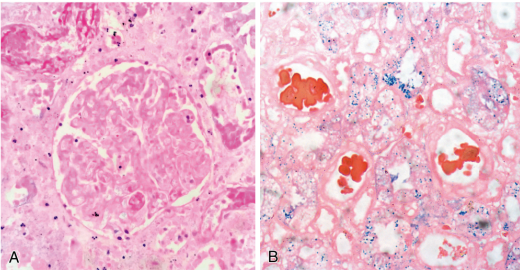


Fig. 59.6 Malaria. (A) Ghost outlines of glomerular and adjoining tubulointerstitial compartment due to acute cortical necrosis in a patient with malarial acute kidney injury (hematoxylin-eosin stain, 40x). (B) Kidney biopsy of a patient with malaria reveals clumps of degenerated red blood cells forming casts with acute tubular injury. Coarse iron pigment is seen in the cytoplasm of proximal tubules (Prussian blue stain, 20x). (Image courtesy Dr. Mahesha Vankalakunti.)

Fig_59_6

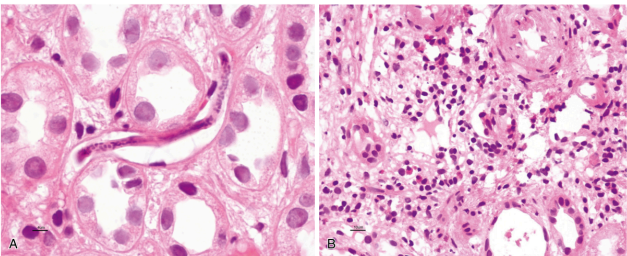


Fig. 59.7 (A) Photomicrograph showing larva of microfilaria in the peritubular capillary lumen in a patient with acute kidney injury (hematoxylin-eosin [H&E] stain, 100x). (B) Acute interstitial edema and inflammatory cell infiltrate rich in eosinophils, as an allergic manifestation of filariasis (H&E stain, 40x). (Image courtesy Dr. Mahesha Vankalakunti.)

Fig_59_7

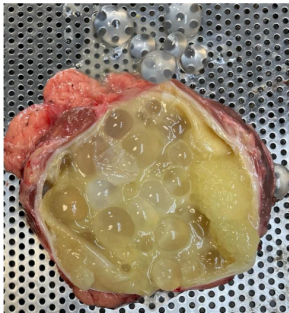


Fig. 59.8 Gross image of the resected kidney (partial nephrectomy) showing a hydatid cyst containing multiple "daughter" cysts. (Image courtesy Dr. Abhishek Singh, Muljibhai Patel Urological Hospital, Nadiad, Gujarat, India.)

Fig_59_8

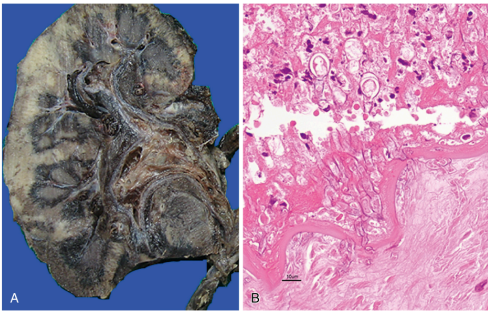


Fig. 59.9 (A) Gross photograph of a nephrectomy specimen showing diffuse hemorrhagic and dark appearance due to mucormycosis. (B) Broad aseptate hyphae of mucormycosis evoking neutrophilic response and invading the internal elastic lamina of the arterial wall (hematoxylin-eosin stain, 20x). (Image courtesy Dr. Mahesha Vankalakunti.)

Fig_59_9

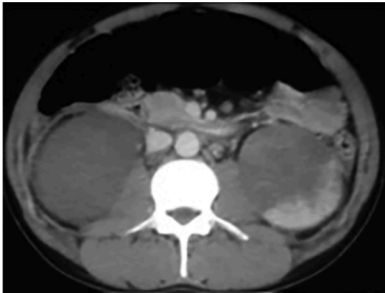


Fig. 59.10 Contrast-enhanced computed tomography scan showing complete nonenhancement of both kidneys due to renal mucormycosis, except part of the left kidney. The plane between the left kidney and the psoas muscle is obliterated, suggesting invasion.

Fig_59_10

Table 59.1 Common Differential Diagnosis of Tropical Acute Febrile Illness Associated with Acute Kidney Injury	
Clinical Picture	Differential Diagnosis
Fever + jaundice	Leptospirosis, malaria, dengue, hantavirus, rickettsiosis, acute hepatitis, Leptospirosis
Biphasic fever + conjunctival suffusion + thrombocytopenia + transaminitis	Hantavirus
Continuous fever + severe respiratory symptoms leading to ARDS	Dengue
Fever + severe myalgia + thrombocytopenia + acalculous cholecystitis	Scrub typhus
Fever + maculopapular rash + eschar	Malaria
Fever + splenomegaly + thrombocytopenia	Brucellosis
Fever + exposure to unpasteurized milk products	Bacterial or viral gastroenteritis
Fever + diarrhea	
AKI, acute kidney injury; ARDS, acute respiratory distress syndrome; CA-AKI, community-acquired acute kidney injury. From Burdman EA, Jha V. Acute kidney injury due to tropical infectious diseases and animal venoms: a tale of 2 continents. Kidney Int. 2017;91(5): 1033–1046.	

Table_59_1

Table 59.2 Characteristics Features of Various Infections with Associated Renal Syndromes						
Infections	Causative Agent	Geographic Areas	Clinical Characteristics	Renal Syndromes	Renal Histopathology	Treatment
Bacterial Infections						
Lepnemy	Mycobacterium leprae	Brazil, Africa, and southern Asia	Hypopigmented skin lesions on the face, diminished or complete loss of sensation with localized areas of the skin, and paronychia	Concentrating defects, AKI, CKD, nephrotic syndrome, and GN	ATN, Proliferative GN, MPGN, MPO, amyloidosis, glomerulonephritis, and GN	Multidrug therapy with rifampin, clofazimine and clarithromycin
Enteric fever	Salmonella typhi	Africa, Eastern Mediterranean, Southeast Asia, and Western Pacific Regions	Fever, abdominal pain, loss of appetite, and diarrhea	AKI and HUS	Acute necrotic GN, crescentic GN, glomerulonephritis, ATN, interstitial nephritis, and crescentic GN	Fluoroquinolones, azithromycin, and cephalosporins
Shub typhus	Orientia tsutsugamushi	Southeast Asia, India, China, Japan, Indonesia, and northern Australia	Fever, eschar, myalgia, lymphadenopathy, headache, gastrointestinal symptoms	AKI, proteinuria, nephrotic syndrome, and hemolytic uremic syndrome	ATN, AKI, MPO, and mesangiocapillary glomerulonephritis with mesangial IgA deposits	Antibiotics and supportive care
Leptospirosis	Spirochetes from the genus Leptospira. Bacteria are the primary reservoirs responsible for transmission	South and Southeast Asia, Oceania, the Caribbean, sub-Saharan Africa, and Latin America	Fever, jaundice, myalgia, bleeding diathesis, conjunctival hyperemia, gastrointestinal symptoms, and hepatomegaly	Fever, jaundice, myalgia, bleeding diathesis, conjunctival hyperemia, and tubular disorders, and vasculitis	ATN, AKI, mesangial glomerulonephritis, and vasculitis	Cephalosporins, doxycycline, and azithromycin
Viral Infections						
COVID-19	SARS-CoV-2	Pandemic	Cough, muscle pain, fever, dyspnea, sore throat, fatigue, headache, and anorexia	Proteinuria, hematuria, renal impairment, and acute kidney injury	ATN, podocytopathy, COVID-19, glomerular disease, AKI, CKD, immune-mediated	For the severity (see text)
Dengue fever	Dengue virus. Primary vector is Aedes aegypti mosquito	Africa, the Americas, the Eastern Mediterranean, Southeast Asia, and the Western Pacific	Headache, retro-orbital pain, fever, arthralgia, myalgia, and thrombocytopenia, increased hematocrit and leukopenia	Proteinuria, AKI, nephrosis, and HUS	Proteinuria, AKI, nephrosis, and HUS	Fluids and conservative management
Parasitic Infections						
Malaria	Caused by Plasmodium. Transmitted via female Anopheles mosquito	Sub-Saharan Africa, South America, Southeast Asia, Eastern Mediterranean, Western Pacific, and the Americas	Fever, accompanied by other symptoms such as chills and rigors, headache, weakness, gastrointestinal discomfort, muscle pain, thrombocytopenia, and hepatosplenomegaly	AKI, hematuria, HUS, nephrosis, ESRD	ATN, AKI, collapsing glomerulopathy, MPO, AKI, immune-mediated proliferative GN, or TMA	Antemalarials, supportive, and quinine

Table_59_2_part1

Table 59.2 Characteristics Features of Various Infections with Associated Renal Syndromes—cont'd						
Infections	Causative Agent	Geographic Areas	Clinical Characteristics	Renal Syndromes	Renal Histopathology	Treatment
Leishmaniasis	Caused by various protozoan parasites of the genus Leishmania	Asia, Africa, the Americas, and Mediterranean region	Fever, malaise, weight loss, and splenomegaly	Proteinuria and hematuria, renal tubular dysfunction and AKI	AKI, proteinuria, and necrotizing glomerulonephritis and amyloidosis	Amphotericin B, paracetamol, antifungal medicines, and melarsoprol
Flavivirus	Bushveldt fever, Dengue fever, and Zika virus. Main vector: Aedes mosquito	Sub-Saharan Africa, Southeast Asia, the Indian subcontinent, the Pacific Islands, and specific regions in Latin America and the Caribbean	Acute febrile illness, myalgia, headache, conjunctivitis, and respiratory symptoms	Chyluria, microscopic hematuria, proteinuria, nephrotic syndrome, and AKI	Proliferative GN, MPO, and renal amyloidosis	Supportive care, and antiviral medicines
Echinococcosis	Caused by various species of Echinococcus. Dogs are definitive hosts. Ungulates, including sheep and goats, are intermediate hosts, and humans are accidental intermediate hosts	Central Asia, western China, South America, eastern Africa, and Mediterranean countries	Asymptomatic, fever, dull flank pain, nausea, vomiting, and typhoid	Proteinuria and abdominal masses that resemble tumors	Cystic degeneration of MPO, MPO, and amyloidosis	Surgical, albendazole
Fungal Infections						
Candidiasis	Candida species, such as C. albicans, C. glabrata, C. parapsilosis, C. lusitana, and C. tropicalis	Mostly worldwide in individuals with compromised immune systems. Other predisposing factors include prolonged ICU stay, use of catheters, urinary drainage devices, diabetes, and urinary tract abnormalities	Fever, flank pain, rigors, chills, or passage of purulent material and painless rigors (the presence of a large ball)	AKI and pyuria	Fungal glomerulonephritis, interstitial nephritis, necrotizing, renal infarction, and renal papillary necrosis	Fluconazole, amphotericin B
Mucormycosis	Most infections are caused by Rhizopus spp.	Mostly occur in immunocompromised individuals, especially those with COVID-19 infection, trauma, burns, debilitation, therapy for immunosuppression, and intravenous drug use	Fever, flank pain and tenderness, other systemic involvement such as cutaneous, rhinocerebral, pulmonary, and gastrointestinal can be there	Gross hematuria, pyuria, oliguria or AKI	Intraluminal abscesses. Vascular thrombosis can cause ischemia and necrosis of kidney tissue. Fungi may infuse the perinephrium, glomeruli, and tubules	Debridement of involved tissue, amphotericin
Aspergillitis	Invasive cases are mostly attributed to species of Aspergillus. Fungal complex	Common in immunocompromised patients. Can occur in patients with genitourinary tract infection	Flank pain, fever, and systemic involvement such as weight loss and exhaustion	Pyuria, hematuria, and AKI	Pyelitis, necrosis, renal infarction, and renal abscesses	Vincristine, amphotericin B, and surgical debridement

AKI, Acute kidney injury; ARDS, acute respiratory distress syndrome; CA-AKI, community-acquired acute kidney injury; COVID-19, coronavirus disease 2019; CKD, chronic kidney disease; COVID-19, coronavirus disease 2019; ESRD, end-stage renal disease; GN, glomerulonephritis; HUS, hemolytic uremic syndrome; ICU, intensive care unit; MPO, mesangiocapillary glomerulonephritis; MPO, mesangiocapillary glomerulonephritis; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; TMA, thrombotic microangiopathy.

Table_59_2_part2